MIL-I-12714 (ORD)
13 May 1953
SUPERSEDING
Army 51-72-18
12 May 1937

MILITARY SPECIFICATION

INSTRUMENT, OBSERVATION, A.A., B.C., M1
(WITHOUT TELESCOPES AND TRIPOD)

This specification consists of this cover sheet and Specification 51-72-18, dated 12 May 1937, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Custodian:
Army - Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:13Z
Check the source to verify that this is the current version before use.

U. S. ARMY
SPECIFICATION

No. 51-72-18

MAY 12, 1937

INSTRUMENT, OBSERVATION, A. A., B. C., M1 (WITHOUT TELESCOPES AND TRIPOD)

A. APPLICABLE SPECIFICATIONS.

A–1. Ordnance Department Drawing, class 22, division 187, drawing 1, titled "Instrument, Observation, A. A., B. C., M1", lists all drawings and specifications necessary to complete the manufacture of this instrument.

A–2. All drawings and specifications listed on the above drawing, of the issue in effect on the date of invitation for bids form part of this specification.

B. TYPES.

B–1. This specification covers one type of instrument known as Instrument, Observation, A. A., B. C., M1. It also covers all items of equipment shown on the drawings as forming parts of the complete instrument except the following:

	U. S. Army specification
Telescope, elbow, M2	51–72–24
Telescope, elbow, M2A1	51–72–24
Telescope, M3	51–72–23
Tripod, type A	51–78–15

B–2. The telescopic equipment regularly employed on this instrument consists of one Telescope, M3, with one Telescope, Elbow, M2, assembled thereto. In certain cases, however, one additional telescope, the Telescope, Elbow, M2A1, is employed assembled to the Telescope, M3, by a special adapter. See par. H–1.

C. MATERIAL AND WORKMANSHIP.

C–1. Material and workmanship shall be in accordance with the drawings, general specifications, and material specifications forming parts of this specification.

C–2. Components and material of commercial nature, for which no requirements are contained herein, or for which no specifications are listed on the list of specification drawing, shall be of the type most suitable for the purpose, and shall be of the best commercial quality, unless a lower grade is specified.

C–3. In order to obtain the accuracy and performance required in section E, special attention is necessary to the following:

C–3a. *Caps, ball, and sockets, ball.* (*A32118, A32123, A32124, A34054*).—Concentricity of exterior surfaces with center axis of spherical surfaces for worm bearings.

C–3b. *Plungers, worm* (*A37208 and 38280*).—Intersection of axis of slot with longitudinal axis of plunger.

C–3c. *Micrometer, A38728; Micrometer, A32104; Adapter, Micrometer, 35719; Ball, Leveling, A38249.*—Concentricity of turned and bored surfaces.

Source: https://assist.dla.mil -- Downloaded: 202
Check the source to verify that this is the current v

C–3d. *Worm (B129181 and B16312).*—1. Concentricity of bearing surfaces and pitch circle helix with axis of worm.

2. Uniformity of lead and smoothness of threads.

C–3e. *Housing (D–9958).*—1. Concentricity of 8.625 inch diameter with 2.8125 inch tapered bearing.

2. Location of axis of bored hole for worm in plane perpendicular to axis of 2.8125 inch bearing.

3. Perpendicularity of horizontal bearing surfaces for Yoke, D9860, and Washer, Retaining, A38264, to axis of tapered bearing.

4. Location of axis of 0.625 inch hole for plunger in same plane with axis of hole for worm.

C–3f. *Housing (C–44209).*—1. Parallelism of axis of hole for worm with side bearing surfaces for worm wheel segment, C44568, and with surfaces of mounting pads.

2. Location of axis of plunger holes in plane parallel to side bearing surfaces for worm wheel segment.

3. Fit of bearing to worm wheel segment (C44568).

C–3g. *Wheel, Worm (C44567).*—1. Concentricity of pitch circle with axis of tapered spindle.

2. Uniformity of pitch of teeth.

3. Perpendicularity of plane of pitch circle with axis of tapered spindle.

4. Concentricity of ¾–16NF–3 threads for ball, leveling, A38249, with tapered spindle.

5. Lapping tapered spindle to Yoke, D9860, to obtain bearing on both spindle and Washer, Yoke, A38266, without perceptible looseness.

C–3h. *Segment, worm wheel (C44568).*—1. Location of center of pitch circle.

2. Parallelism of pitch circle to side bearing surfaces.

3. Uniformity of pitch of teeth.

4. Fit of guides to Yoke, D9860, and Housing, C44209.

C–3i. *Yoke (D9860).*—1. Concentricity of turned diameters of base and tapered hole.

2. Perpendicularity of axis of bearings for telescope with axis of tapered hole.

3. Perpendicularity of turned shoulders with axis.

4. Location of axis of telescope bearings in plane of axis of tapered hole.

5. Finishing bosses for housing, C44209, parallel to axis of tapered hole.

6. Lapping to worm wheel (C44567) as stated in par. C–3g 5.

D. GENERAL REQUIREMENTS.

D–1. See U. S. Army Specification No. 51–7, "General Specification Governing the Manufacture and Inspection of Fire Control Instruments."

E. DETAIL REQUIREMENTS.

E–1. *Interchangeability* shall be as defined in the paragraph on interchangeability in Specification No. 51–7, "General Specification Governing the Manufacture and Inspection of Fire Control Instruments."

[No. 51–72–18]

Source: https://assist.dla.mil -- Downloaded: 2026-10-1
Check the source to verify that this is the current version

E–2. *Accuracy and adjustment.*—

E–2a. All moving parts shall function properly, without irregularities, undue looseness, or friction.

E–2b. The levels shall be adjusted to show the bubbles central with respect to their graduations when the axis of the worm wheel is in a vertical position.

E–2c. The azimuth micrometer shall be adjusted to indicate zero when any 100 mil graduation of the azimuth scale is opposite the azimuth index, with the azimuth worm in mesh.

E–2d. The elevation index shall be adjusted to indicate zero when the instrument is leveled as in E–2b above, and the line of sight of a telescope mounted in the instrument is horizontal.

E–2e. The elevation micrometer shall be adjusted to read zero when any 100 mil graduation of the elevation scale is opposite the elevation scale index.

E–2f. Electrical equipment shall function properly as intended so that the lines on the reticle, and readings at the scales and micrometers will be clearly visible, without glare when tested in a dark room.

E–2g. The position of the level bubbles shall not vary more than 45 seconds from the central position, as described in E–2b above, when the yoke is rotated through 360° with the rest of the mount stationary.

E–2h. With the instrument leveled, and with the telescope mounted in place, the line of sight of the telescope shall not deviate from a true vertical line more than 0.5 mil when the telescope is rotated in the trunnion bearings from 0 to 800 mils, and not more than 0.75 mil when elevated to 1,500 mils.

E–2i. Backlash in the elevation mechanism shall not exceed 0.5 mil.

E–2j. Backlash in the azimuth mechanism shall not exceed 0.3 mil.

E–2k. Error of any azimuth reading in 360 shall not exceed plus or minus 0.75 mil.

E–2l. Error of any elevation reading from 0 to 800 shall not exceed plus or minus 1 mil, and from 800 to 1,500 the error shall not exceed 1¼ mils.

E–3. *Lubrication* shall be in accordance with drawings and specifications 51–7, "General Specification Governing the Manufacture and Inspection of Fire Control Instruments."

E–4. *Painting and finishing* shall be in accordance with Drawings and Specification No. 51–70–1, "General Specification Governing Painting and Finishing of Fire Control Instruments."

E–5. *Manufacture of extra parts* shall be in accordance with this specification.

E–6. *Serial numbers* will be furnished by the procuring office, and shall be inscribed in the manner shown on the drawings.

F. METHODS OF INSPECTION, TESTS, ETC.

F–1. Each instrument and component parts thereof, including extra parts, shall be inspected during process of manufacture and assembly, as provided by Specification No. 51–7, "General Specification Governing the Manufacture and Inspection of Fire Control Instruments."

[No. 51–72–18]

Source: https://assist.dla.mil -- Downloa
Check the source to verify that this is the

F–2. Each completed instrument shall be tested for conformance with the requirements in section E.

F–3. Each instrument shall be inspected for accuracy of azimuth readings at at least 16 different settings throughout the 6,400 mils of azimuth and for the accuracy of the elevation readings at at least 6 settings, throughout the limits of the elevation scale. The settings shall be selected by the inspector.

F–4. In cases where telescopes and/or tripods are not to be furnished by the contractor with the instruments, a sufficient number of these items shall be furnished by the War Department to permit uninterrupted inspection and test of the instruments.

G. PACKING AND MARKING.

G–1. *Packing.*—

G–1a. Unless otherwise specified, each observation instrument shall be packed for shipment in the chest shown on drawing D9884. The equipment listed in the table of contents on drawing D9884 shall also be packed on the instrument in the chest. The chests, with contents, shall then be packed in substantial wooden packing boxes in quantities specified by the contracting officer.

G–1b. The Battery, Storage, Alkaline, M1, shown on drawing C–44117, shall be packed for shipment in the Box, Battery, Type B4, shown on drawing D9654. The Box, Battery, Type B4, complete with contents shown on drawing D9654, shall then be packed in a substantial wooden box similar to the usual commercial type, designed so that the battery will be maintained in an upright position in handling and shipping, in order to avoid spilling the electrolyte.

G–1c. All packing shall be such as to insure acceptance by common or other carriers for safe transportation at lowest rate to point of delivery.

G–2. Marking shall be in accordance with U. S. Army Specification No. 100–2, "Standard Specification for Marking Shipments."

H. NOTES.

H–1. The instrument, Observation, A. A., B. C., M1, complete, is assembled in one of the following manners:

H1–a. *Instruments equipped with one tracking telescope* include:

	U. S. Army specification
1 Instrument, Observation, A. A., B. C., M1	51–72–18
1 Telescope, M3 (with adapter assembly C–44783)	51–72–23
1 Telescope, Elbow, M2	51–72–24
1 Tripod, type A	51–78–15

H–1b. *Instruments equipped with two tracking telescopes* include:

	U. S. Army specification
1 Instrument, Observation, A. A., B. C., M1	51–72–18
1 Telescope, M3 (with adapter assembly C–44783)	51–72–23
1 Telescope, Elbow, M2	51–72–24
1 Telescope, Elbow, M2A1	51–72–24
1 Tripod, type A	51–78–15

Source: https://assist.dla.mil -- Down
Check the source to verify that this is t

H–1c. When the procurement of a complete instrument, equipped in either manner, is to be made, all specifications shown in H–1a or H–1b, as proper, are necessary, including all supplementary drawings and specifications.

H–2. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

NOTICE.—Whenever Government drawings, specifications, or other data, are used for any purpose other than in connection with a definitely related Government procurement operation, the United States thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder, or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

NOTE.—Copies of this specification may be obtained from the office, Chief of Ordnance, Washington, D. C.

[No. 51–72–18]

Source: https://assist.dla.mil -- Downloade
Check the source to verify that this is the cu